



Q1: [30] Consider an infinitely long, straight cylindrical conductor with radius a , carrying a steady current I . The current is non-uniformly distributed such that the current density $\mathbf{J}$ is proportional to the radial distance ρ from the z -axis. Assume $\mathbf{J} = K\rho\mathbf{a}_z$ for $\rho \leq a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

- (a) [5] Given a and I , what is the value of K ?
- (b) [5+5] Using Ampere's law, determine the magnetic field intensity $\mathbf{H}$ both **inside** and **outside** the conductor.
- (c) [5] Plot the magnitude of $\mathbf{H}$ to show its variation as a function of ρ .
- (d) [5+5] Find the curl of $\mathbf{H}$ to establish that

$$\nabla \times \mathbf{H} = \mathbf{J}$$

is true **everywhere**, i.e., both inside and outside the cable.

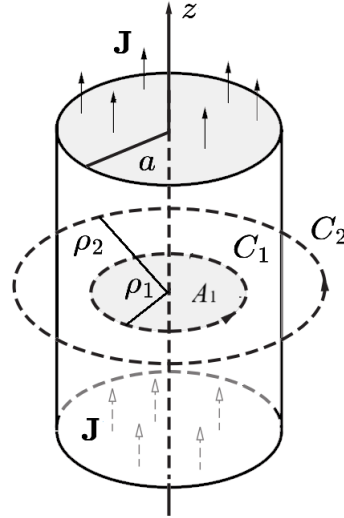


Figure 1: When using Ampere's law, one can consider the circular Ampere contours C_1 and C_2 with radii ρ_1 and ρ_2 , respectively, where $0 < \rho_1 < a$ and $a < \rho_2 < \infty$, to calculate the magnetic field intensity both inside and outside the conductor.

Q2: [25] Consider an infinitely long, straight cylindrical conductor with radius a , carrying a steady current I . The current is non-uniformly distributed such that the current density $\mathbf{J}$ is proportional to the square root of the radial distance ρ from the z -axis. Assume $\mathbf{J} = K\sqrt{\rho}\mathbf{a}_z$ for $\rho \leq a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

- (a) [5] Given a and I , what is the value of K ?
- (b) [5+5] Find $\mathbf{A}$ inside the conductor ($\rho < a$).
- (c) [5+5] Find $\mathbf{A}$ outside the conductor ($\rho > a$).

Hint: Obtain $\mathbf{B}$ using Ampere's law or Biot-Savart's law, and solve $\mathbf{B} = \nabla \times \mathbf{A}$ to find the magnetic vector potential $\mathbf{A}$.

Q3: [15] Consider a rectangular loop placed near a pair of infinitely long wires, cables 1 and 2, as depicted in Figure 2. The two wires carry currents I flowing in opposite directions, with the magnitude of the current increasing with time, $I(t) = Kt$, $K > 0$.

- (a) [10] Determine the electromotive force (V_{ind}) induced in the loop (i.e., the potential difference measured across the small gap if the loop were broken) for the rectangular loop shown in Fig. 2.
- (b) [05] Explain why the direction of the induced current is as shown in the figure and not in the opposite direction. Provide a detailed explanation, considering the roles of both cables 1 and 2.

Answer: The induced emf for the circuit is

$$V_{\text{ind}} = \frac{\mu_0 h K}{2\pi} \ln \left[\frac{a(b + a + w)}{(b + a)(a + w)} \right]$$

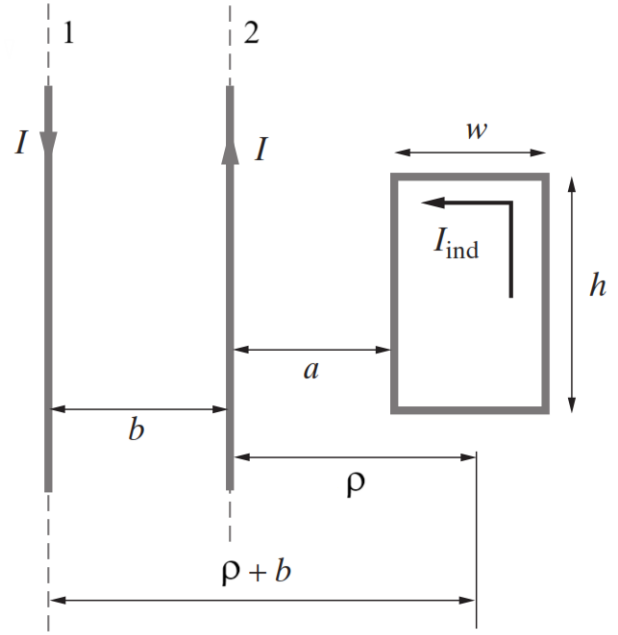


Figure 2: Pair of parallel wires and a loop.

Q4: [30] Upon completing this course, you may consider applying your electromagnetics knowledge to real-world scenarios. Imagine a large power line passing near your locality, with your property 20 meters away from the line at ground level. You have 200 meters of copper wire available, which can be fashioned into a single-turn rectangular loop with dimensions a and b , as shown in Fig. 3. Assume that the power line carries a sinusoidal current with an amplitude of $I_0 = 4000$ A and a frequency of $f = 50$ Hz. The instantaneous current is given by $I(t) = I_0 \cos(2\pi ft)$.

[Caution: Do not try this at home.]

- (a) [10] Considering the fixed perimeter of the loop: $l = 2a + 2b = 200$ m, show that the flux passing through the loop is:

$$\Psi(t) = \frac{\mu_0 b I_0 \cos(2\pi ft)}{2\pi} \ln \left(\frac{20 + a}{20} \right)$$

- (b) [05] Using Faraday's law, show that as

$$V_{\text{ind}}(t) = V_0 \sin(2\pi ft)$$

- (c) [05] Find V_0 , the amplitude of induced voltage, for $b = 10, 20, 40, 60, 80$ and 95 meters.

- (d) [10] Plot V_0 versus b , and figure out the optimal value of b up to one decimal space that gives the largest value of V_0 .

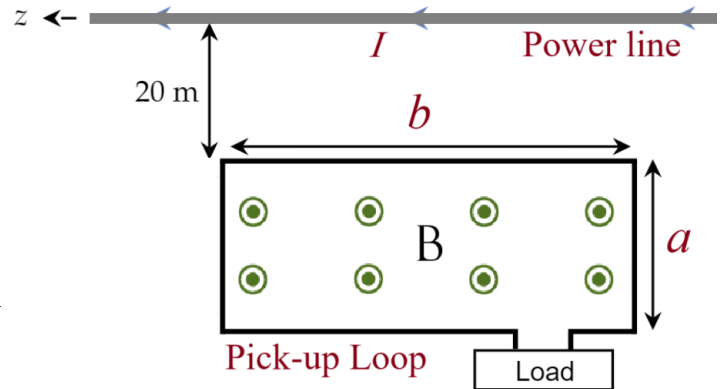


Figure 3: Extracting power from a power line.